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Shopping missions: An analytical method for the identification of shopper need states

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ABSTRACT

Shoppers enter stores to meet diverse need states, such as getting a soft drink and a bag of potato chips for immediate consumption or purchasing ingredients for meals to make on the same day. The present article introduces the concept of a *shopping mission* that can be identified by the complementarity and sales affinity among the balanced number of product categories in trips with relatively concrete goals. The authors develop an analytical method to identify several shopping missions at the store level and demonstrate the utility, validity, and replicability of this method using a data set with 4 million baskets from a multinational supermarket chain. The authors also provide evidence that the identification of shopping missions offers a wealth of opportunities for targeted shopper marketing activities.

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1. Introduction

The importance of shopper marketing, that is, the planning and execution of all marketing activities that influence a shopper along and beyond the entire path to purchase (Shankar, Inman, Mantrala, Kelley, & Rizley, 2011), is clear; both retailers and shoppers cite shopper marketing as having the biggest influence on retail sales (Progressive Grocer, 2014). Shopper marketing literature acknowledges the essential qualities of different types of shopping trips (Walters & Jamil, 2003), ranging from the very precise and concrete (e.g., to take advantage of a specific promotion) to rather abstract (e.g., to shop for the whole week or more) (Bell, Corsten, & Knox, 2011). Relatively concrete shopping trips, such as the purchase of products for immediate or same-day consumption, explicitly signal the reasons shoppers enter stores and provide information valuable for the development of targeted, relevant, and effective shopper marketing activities. Such an approach is already evident to retail practitioners, who regard consumption needs as bases for understanding shopping needs and triggering precise "shopping missions" (ECR Europe, 2011). By understanding these "missions," retailers can develop format- or even store-specific, mission-informed shopper marketing actions, while consumer goods manufacturers can strategically position brands within shopping missions as means to increase relevance and convenience for shoppers. For instance, retailers might be interested to see if consumers purchase for breakfast only as part of major trips, and manufacturers may seek to improve the penetration of a product category in the "meal," rather than just the "breakfast," shopping mission, as well as understand the conditions in which certain trips occur, the link to money spent, and the influences on coupon redemption and pricing decisions.

Prior literature advocates that product categories can provide a basis for identifying latent shopping goals and choices. Shopping trip planning occurs at the category rather than the brand or stock-keeping unit (SKU) level (Block & Morwitz, 1999). Sales data make sales affinities across product categories readily available. The omnipresence of point-of-sale (POS) systems and recent advances in analytical and data processing capabilities for marketing research (Germann, Lilien, & Rangaswamy, 2013) spur the exploitation of sales data for the development of actionable shopper insights (Fox & Sethuraman, 2006; Humby & Hunt, 2003). The identification of shopping missions, that is, the identification of need states as captured in purchasing patterns solely from sales data, gives retailers the possibility, with modest investments, to gain information about a host of targeted shopper marketing activities.

This research suggests a method for identifying shopping missions grounded in the inherent tendencies of the product categories to be consumed and, more important, purchased together. The authors explain the need for a construct such as shopping mission, which they develop conceptually by reviewing its connection to the goals and mindsets of shoppers, as suggested in pertinent literature, and through discussions with 58 shoppers in eight focus group sessions. This article recommends an effective analytical method for the identification of shopping missions by clustering product categories found in grocery sales data and demonstrates its utility with a data set comprised of 4 million baskets from six stores of different formats in a multinational supermarket chain. Shopping missions yield significant insights on

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shopper purchasing behavior that can form the basis for shopper marketing actions driven by personalization, customization, and efficiency.

2 . Theoretical background

2.1 . Shopping goals

Motivational theories of social psychology suggest that shopping goals (Bagozzi & Dholakia, 1999; Bridges & Florsheim, 2008; Lee & Ariely, 2006) provide shoppers with a sense of direction and clarity for their shopping actions, influencing the way they think and behave while shopping (Kruglanski et al., 2002; Locke & Latham, 1990). Prior literature distinguishes between abstract and concrete shopping goals (Lee & Ariely, 2006), suggesting that goals are established before the shopper enters the store and expressed as different shopping trip types (Bell et al., 2011). Shopping trip goals range from very precise and concrete, such as visiting the store for a specific promotion (Mulhern & Padgett, 1995; Walters & Jamil, 2003); to concrete, such as purchasing products for immediate consumption or for meals of the same day (Bell et al., 2011); to relatively abstract fill-in trips aimed at a general replenishment of frequently consumed perishables; to more abstract and less frequent major trips aimed at obtaining a large number of miscellaneous items (Block & Morwitz, 1999; Kollat & Willett, 1967). The different shopping trip goals and levels of abstractness influence important consumer attitudinal and behavioral outcomes, such as perceived similarity among alternatives (Lamberton & Diehl, 2013), persuasion effectiveness (Nenkov, 2012), promotion effectiveness (Walters & Jamil, 2003), and unplanned buying (Bell et al., 2011). Consumers' goal-oriented behavior also consists of phases, including setting and striving toward goals (Bagozzi & Dholakia, 1999). Goal systems theory (Kruglanski et al., 2002) indicates that activating a goal also activates the cognitive procedures for accomplishing the goal. Thus, the psychological processes associated with either the setting or the striving toward shopping goals activates different cognitive procedures while shopping.

2.2 . Mindset theory of action phases

The mindset theory of action phases (Gollwitzer, 1990) refers to particular sets of activated cognitive procedures, which result in particular ways of processing information. Based on the distinction between motivation and volition, Gollwitzer and colleagues suggest that individuals in a predecisional phase activate a deliberative mindset, which helps them assess the pros and cons of alternatives and whether options are desirable or not. In contrast, individuals in the preactional phase activate an implemental mindset, which helps initiate action and apply instruments to implement the goal at hand. On the basis of recent studies that posit that the point of the shopping course influences a shopping mindset (Dhar, Huber, & Khan, 2007; Lee & Ariely, 2006), but so do preshopping factors such as consumers' chronic shopping orientation (Büttner, Florack, & Göritz, 2013), this research argues that shopping mindset is a physiological process contingent on the overall shopping trip goal assumed by the shopper prior to entering the store.

Shoppers engaged in abstract shopping trips seek to make the majority of their product category decisions as early as possible in the shopping process (e.g., before entering the store). Abstract shopping trips require more time and effort by the shopper, who plans to purchase a large number of items and categories. Seeking to minimize the time spent shopping and safeguard against unplanned purchases or budget deviations, shoppers in major and fill-in trips often use external memory aids, such as shopping lists that help shoppers implement a large shopping task. Similarly, the more focused and selective orientation of an implemental mindset better describes the cognitive and behavioral procedures of shoppers in trips aimed at a handful of categories, such as for special offers (Mulhern & Padgett, 1995) or “cherry-picking” (Talukdar, Gauri, & Grewal, 2010).

When undertaking relatively concrete shopping trips, targeted at short-term consumption occasions or special events, shoppers are aware of the explicit and discrete need(s) they seek to meet (e.g., quick lunch solution, tonight's dinner with guest), construe the products to satisfy their needs tacitly, and deliberate among alternative product solutions (e.g., between pasta salad and pizza for lunch, between white and red wine to complement tonight's dinner) while shopping. Deliberating among alternatives activates cognitive procedures that facilitate preference setting. Shoppers in relatively concrete shopping trips, not knowing yet in which direction their product decisions will take them, seek stimulation in the store, such as by browsing around or discovering and evaluating new products. Therefore, their information processing should be tuned toward deliberating about the pros and cons of products and thinking about wishes and desires. These consumers likely shop with a deliberative mindset.

2.3 . Shopping mission

Shoppers acknowledge the product-related reasons for a store visit by making explicit references to their intended purchases as distinct product entities, often summarized at the product category level in a shopping list for reference and guidance. This tactic is particularly common for shopping trips under an implemental shopping mindset, in which shoppers explicitly acknowledge their purchases prior to entering the store. In contrast, shoppers under deliberative mindsets construe their purchases implicitly and make their category choices in the store. Product categories appear as merchandise organizational units, containing competing product brands, and also can serve the purpose of describing shopping objectives of trips under an implemental shopping mindset. However, a higher-order, cross-category merchandise construct for competing product categories is necessary to designate the shopping objectives of trips under a deliberative shopping mindset. Prior research stresses the importance of higher-order, goal-based interrelationships among products spanning merchandise categories in areas such as marketing communications (Englis & Solomon, 1996) and consumption stereotypes (Chaplin & Lowrey, 2010). In contrast with previously proposed constructs—such as transaction size in value (Kollat & Willett, 1967), number of items purchased (Bell & Lattin, 1998), use or absence of shopping list, overall shopping trip, and store specific goals (Bell et al., 2011)—that distinguish shopping trips on the basis of non-product dimensions, shopping mission—conceptualized as an instantiation of or value assigned to the deliberative shopping mindset—is a product-related categorization of shopping trips that involves the specific need states of shoppers entering stores. Fig. 1 presents the conceptual relationship of shopping goals, mindsets, and missions.

Very concrete shopping trips, such as the purchase of special offers, are particularly prone to implemental mindsets and involve a limited number of items, commonly confined within a single product category. The underlying consumption need is typically self-evident, in that it relates to the purchased category's role, particularly for destination categories. Likewise, abstract shopping trips, such as preplanned fill-in and major trips, involve a large assortment of items in heterogeneous product categories that address a host of miscellaneous needs. Shopping missions are thus hard to identify, due to the incongruent and nested relationships among the multitude of product categories involved. However, relative concreteness and the deliberative mindset it brings about make trips involving a balanced number of product categories, such as for purchasing products for immediate or same-day consumption, particularly applicable to the identification of shopping missions. The inherent relationships, in the form of sales affinities and logical complementarities, among the relatively small number of product categories involved explicitly reflect the need states they seek to meet. Table 1 summarizes the key features of different types of shopping trips and how they relate to the identification of shopping missions.

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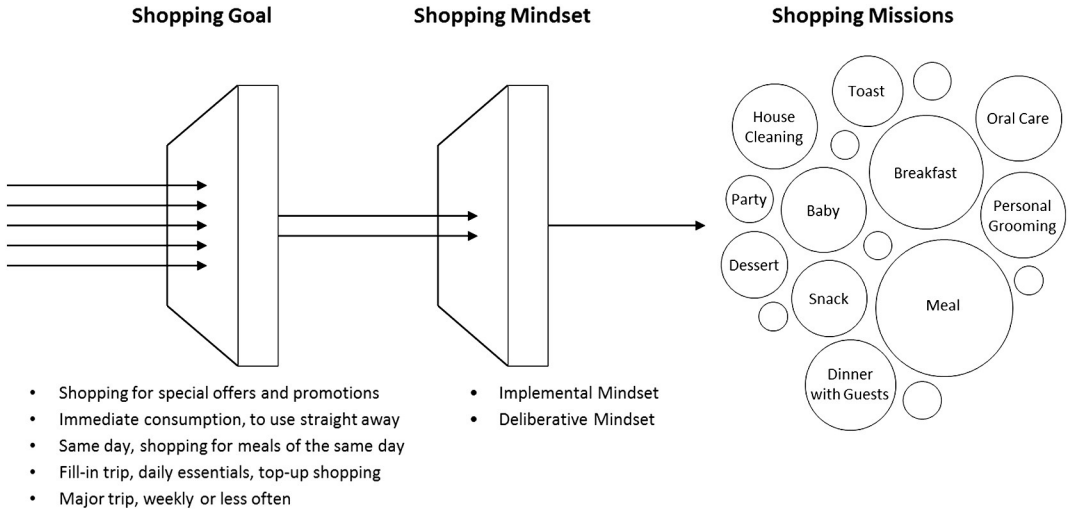


Fig. 1. The conceptual relationship of shopping. Goals, mindsets, and missions.

2.4 . Exploratory qualitative research

To better understand and conceptually define shopping missions and their relationship with different shopping goals and mindsets, the authors conducted a series of discussions with 58 shoppers in eight focus group sessions. Discussants were randomly contacted via telephone and, after an initial screening, offered a voucher for their participation. Each session involved 7–8 shoppers and lasted 45 min. Participants provided consent to videotape the discussions, and all recordings were subsequently transcribed. The shoppers connected shopping trips of modest size and concreteness to personal nutrition and household needs, such as shopping for products for breakfast, dinner, and at-work snacking. Such shopping missions represent need states, usually performed on weekdays and at a relatively small temporal distance from the consumption of the products (same day or day after). As a male shopper in his early 40s notes,

> "I often go grocery shopping on the way home from work. I usually look for a meal solution for my dinner later that day or something to have for lunch the day after."

With respect to such visits, shoppers agree that though they have established a shopping mission prior to entering the store, they decide on alternative product solutions while shopping. Shoppers go further to suggest that when in the midst of such trips, they are particularly open in recommendations intended to complement their need states. In the words of a female shopper in her early 30s,

> "During my lunch break I regularly go to the store nearby without knowing what to buy.… Most of the times I am looking for an 'inspiration' and I am open to any healthy and tasty suggestions I might get."

Shoppers cite convenience and smaller store formats as preferable for fulfilling shopping missions such as breakfast, at-work snacking, and same-day meals, whereas larger establishments such as hypermarket stores are favored for shopping missions relating to household cleaning and personal hygiene. These relationships reflect, among other things, the parking facilities that larger stores offer, which make shopping for heavy packaged goods to meet household cleaning needs more convenient.

3 . Analytical shopper marketing research

Recent technological advances have shaped retailing into a rapidly evolving, technology-intensive, and data-rich industry and profoundly affected the feasibility for conducting shopper marketing research. The ease with which new analytical techniques can discover patterns in large data sets offers great utility and value, such that "We need to avoid the tendency to downplay such methods [data mining and machine learning] as purely data driven and atheoretical. This is not to say that there is no value in theory, *but these methods are helpful to identify new phenomena*, new perspectives, and new theory" (Gupta, Hanssens, Hauser, Lehmann, & Schmitt, 2014, p. 4, emphasis added).

Prior research employs grocery scanner data to investigate consumer behavioral reactions to price promotions (Mayhew & Winer, 1992), brand choice (Chib, Seetharaman, & Strijnev, 2014), and cross-category price elasticities (Russell & Petersen, 2000), as well as to segment shoppers on the basis of basket size (Bell & Lattin, 1998) or

Table 1
The identification of shopping missions contingent upon abstractness and mindsets of shopping goals.

Abstractness	Predominant mindset	Exemplar shopping goals	Authors	Underlying consumption need(s)	Identification of shopping mission(s)
Veryconcrete	Implemental	Shopping for special offers and promotions	Mulhern and Padgett (1995)Walters and Jamil (2003)Bell et al. (2011)	Self-evident,related to the (single) product category(-ies) involved	Yes, in case destination category(-ies) involved
Relatively concrete	Deliberative	Immediate consumption, to use straight awaySame day, shopping for meals of the same day	Bell et al. (2011)	Explicit/ distinct	Yes, due to the inherent relationships (sales affinity, logical complementarity) of the small number of product categories involved
Relatively abstract Veryabstract	Implemental	Fill-in trip, daily essentials, top-up shoppingMajor trip, weekly or less often	Kollat and Willett (1967)Mulhern and Padgett (1995)Block and Morwitz (1999)Walters and Jamil (2003)Bell et al. (2011)	Several/ diverse	Hard, due to the incongruent and nested relationships among the large number of product categories involved

other variables such as trip frequency and wallet share (Talukdar et al., 2010). Professional market research firms similarly rely on grocery sales data to provide insights to manufacturers and retailers of consumer packaged goods. However, insights on trip-level shopper goals and objectives remain confined to understanding *how* consumers shop, by segmenting shopping trips on the conceptual grounds of overall shopping trip abstractness (Brown, 2010; IRI, 2006). The identification of shopping missions goes beyond the segmentation of trip types, to the discovery of constellations of complementary grocery product categories whose sales affinity forms the basis for understanding specific "need states" or *why* a shopper visits the store.

As (shopper) marketing and information technology become increasingly interwoven, the design science paradigm (Hevner, March, Park, & Ram, 2004) offers an informative and technology-active research approach for marketing. Design science research focuses on creating and evaluating innovative artifacts (such as tools and methods) that enable researchers to identify new phenomena of interest. The design of new methods that uncover potential phenomena can complement more traditional methods, and then used to test and elaborate the theories (Gupta et al., 2014). Analytical data-driven techniques methodologically rely on statistical learning (Hastie, Tibshirani, & Friedman, 2009), that is, the partitioning of large amounts of empirical observations into training and testing sets to evaluate the models and conclusions induced from observations. The current work contributes to growing literature that uses statistical learning techniques to generate shopper insights (Cil, 2012; Oestreicher-Singer, Libai, Sivan, Carmi, & Yassin, 2013; Papakiriakopoulos, Pramatari, & Doukidis, 2009) while also responding to recent calls to identify new phenomena of interest for scholarly marketing research.

4 . Methods and data

This article suggests a data-driven method to identify shopping missions by clustering basket data, according to the product categories included. At the heart of the proposed method is the characterization of each basket on the basis of the categories it includes. The method consists of three data management phases, namely, sales data preparation, category taxonomy preparation, and sales-category consolidation. Two analysis phases follow, namely, shopping goal segmentation and the identification of shopping mission phases. Fig. 2 illustrates the proposed method and constituent phases. The following sections present the goal of each phase, along with a discussion of any issues expected to emerge at each point.

The demonstration of the utility of the proposed method relies on a sales data set from a multinational retailer operating in Greece. To avoid any seasonality effects, the sales data reflect 17 consecutive months, January 2012 to May 2013. Furthermore, the analysis focuses on three pilot stores with different formats: a small convenience store, a supermarket, and a hypermarket, with 220,1000, and 12,000 square meters of selling space, respectively. Examining stores of different sizes helps confirm the validity of the proposed approach, showing that the resulting clusters relate to the external variables in ways anticipated by the theoretical development, previous research, and exploratory discussions. This study also includes three control stores with similar sizes, product assortments, and clientele geodemographic properties, to attest to the consistency and stability of the solution across different conditions. Overall, the sales data set comprises 35,914,213 SKU product-level purchase records from six stores situated in the larger Athens metropolitan area.

Table 2
SKU- and basket-level sales records for all stores.

Store	SKU-level	Basket-level
Hypermarket pilot	15,084,588	1,525,229
Hypermarket control	13,448,929	1,193,077
Supermarket pilot	2,761,610	412,786
Supermarket control	2,765,150	369,158
Convenience pilot	725,562	158,572
Convenience control	582,631	139,208

5 . Analysis and results

5.1 . Sales data preparation

Preparatory steps in the unit quantity field ensured that the variable followed a consistent value format and ruled out any errors caused by operational business processes or systems. The execution of the sales data preparation steps resulted in 35,368,470 individual, SKU-level purchase records (545,745 records omitted). Following the necessary adjustments at the lowermost level of granularity (i.e., SKU level), the unit quantity field can be aggregated to the basket level. This step is necessary because the basket, or individual shopping trip, constitutes the level of analysis for identifying shopping missions. The aggregation of the sales records resulted in 3,798,030 unique baskets for all six stores. Table 2 summarizes the SKU- and basket-level records for the six stores. More fields, such as unique identifiers of the channel in which the purchase occurred, a timestamp to denote the date and time when the purchase occurred, or a unit price field that designates the total line item price paid, could be included to enrich the insightfulness of this method.

5.2. Product taxonomy preparation

This method requires a product taxonomy that includes identifiers and descriptive fields for all product category levels used by the collaborating retailer. All stores in the study use the same product taxonomy. Because product categories constitute the building blocks of shopping missions, important or destination categories that are not available

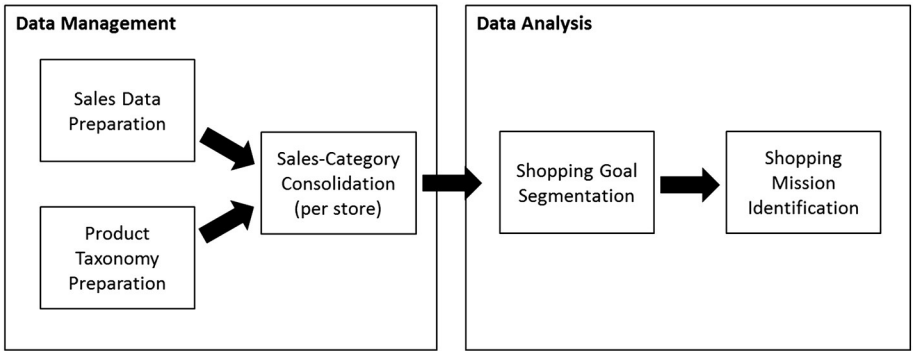


Fig. 2. Phases of the method for identifying shopping missions.

within a store should negatively affect the shopping missions characterized by these categories. For example, if product categories such as milk, detergents, or diapers are absent from a store's product offerings, shopping missions such as breakfast, cleaning, or baby care are negatively affected. Fig. 3 presents the product taxonomy identified, which includes four product category levels with 6, 47, 450, and 1918 unique category values in each level. As retailer-specific merchandise hierarchies, product taxonomies typically demonstrate uneven depths across branches, which make it essential to identify the description level that yields meaningful product category associations and to include product identifiers, at the SKU level, to establish a link between the product taxonomy and the sales data sets. Independent qualitative assessments by two researchers, as well as round-table discussions with retail executives, resulted in 75 product categories available in all the three store formats under consideration.

5.3 . Sales-category consolidation

The basket-level sales data and product category fields that resulted from the previous steps can be combined to create a *fact table* for each of the six stores under consideration, summarizing all available information. Associations among product categories and the resulting shopping missions should differ across stores, due to the varying product assortments, distribution of shopping trip goals, and regional geodemographic properties. All six fact tables contain 78 variable columns: the basket identifier, serving as the unit of analysis; the basket volume and variety fields, serving as descriptive attributes related to shopping trips; and 75 category fields populated with binary flags (participation matrix) to indicate whether a particular basket contains a product identifier belonging to the particular category (1) or not (0).

Table 3
Example participation matrix structure.

Basket ID	Basket volume	Basket variety	Category 1	Category 2	Category 3	…	Category 75
153356567	15	5	1	0	1	…	0
153356588	17	6	1	1	1	…	0
153356898	3	1	0	0	1	…	0
153356903	12	12	0	1	1	…	0
153789632	15	7	0	0	0	…	0
289674955	22	6	0	0	0	…	1
303118876	19	7	1	0	1	…	0
…	…	…	…	…	…		…
Unit of analysis	Shopping trip attributes		Category binary flags				

Table 3 illustrates an example fact table with all available information on the individual basket level for a single store.

The number of observations for each table (table length) corresponds to the number of unique baskets for each store. Table 4 presents and provides descriptive information for all variable fields for each of the six fact tables. Data frames exported from the database management system take the form of .csv or .txt files. Such interoperable file formats can be imported to a host of dedicated statistical analysis software to carry out the subsequent phases of the method.

5.4 . Shopping goal segmentation

Shopping goal segmentation involves a cluster analysis using the basket variety measure, that is, the number of unique product categories within a basket, as a clustering input variable. The analysis thus distinguishes between a customer who is buying many items but just

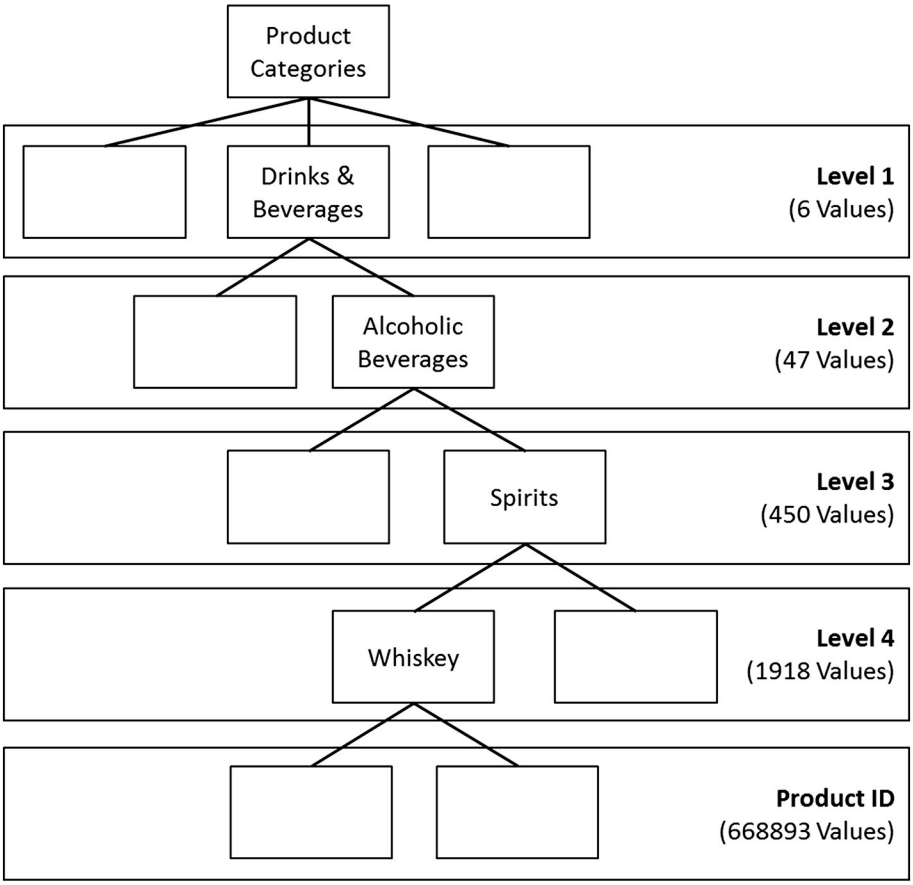


Fig. 3. Product taxonomy structure.

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Table 4
Descriptive statistics for all fact tables.

Variable	Hypermarket		Supermarket		Convenience	
	Pilot	Control	Pilot	Control	Pilot	Control
Basket id[(1)]	1,525,229	1,193,077	412,786	369,158	158,572	139,208
Basket volume[(2)]	24.74	22.56	12.18	11.73	5.77	6.14
Basket variety[(3)]	13.23	13.23	10.12	9.97	5.12	5.27
Category 1[(4)]	31.12%	30.04%	25.16%	26.58%	9.65%	10.32%
Category 2[(5)]	15.63%	15.98%	11.87%	11.52%	10.54%	9.87%
…	…	…	…	…	…	…
Category 75[(6)]	5.66%	5.21%	4.93%	3.78%	6.82%	7.12%

(1) Total observations (rows) of each table.
(2) Mean values (unique items per basket).
(3) Mean values (unique categories per basket).
(4) Percentage of baskets that include at least one product from category 1 (laundry powder soaps).
(5) Percentage of baskets that include at least one product from category 2 (facial creams).
(6) Percentage of baskets that include at least one product from category 75 (coffee and tea).

from one product category (voluminous basket with few categories) and a shopper purchasing the same number of items but from different categories (voluminous basket with a wide variety of product categories). In analytical terms, the Euclidean distance between basket varieties serves as a proximity measure for capturing the (dis)similarity between shopping trips within the k-means partitional cluster analysis for each fact table, with training and testing sets equivalent to 60% and 40% percent of the available observations, respectively. The gap statistic procedure derives the optimal number of clusters (Tibshirani, Walther, & Trevor, 2001). Table 5 summarizes the final cluster solution.

To interpret the cluster analysis results, the basket variety measure serves as an operationalization of shopping goal abstractness. The more (fewer) product categories a shopping trip involves, the more abstract (concrete) its shopping goal. On a per store basis, all baskets can be assigned to three levels of shopping goal abstractness: "concrete," "relatively concrete," and "abstract," according to the basket variety ranges of each resulting cluster. First, clusters 4 and 5 (baskets with the widest variety of categories) combine into an "abstract" group of baskets encompassing a significantly larger number of diverse product categories. Second, clusters 2 and 3 (baskets with a modest variety of categories) group into a "relatively concrete" set of baskets, involving deliberation and satisfaction of short-term, quotidian consumption occasions or special events. Third, cluster 1 (baskets with either a single or few categories) is the group of "concrete" baskets, because the majority of these trips include shopping for special offers, promotions, and similarly targeted grocery visits.

These categories are not to suggest that there are no "abstract" shopping trips, such as fill-in trips for miscellaneous perishables and other daily essentials, with fewer than 16 product categories in the pilot supermarket, or that there are no "relatively concrete" shopping trips, such as assembling the provisions for a picnic or a weekend excursion, with more than 13 product categories in the control convenience store. However, whereas prior literature on shopping trip abstractness elicits shopping trip abstractness directly from shoppers' psychometric self-reports, this cluster analytic framework relies on observed behavioral measures such as basket variety and thus can effectively and efficiently cope with large sales data sets generated by the hundreds of thousands of shopping trips retailers collect and repost daily. Such an approach can provide a precedent and complement to psychometric measurements of shopping goal abstractness, not a replacement.

Furthermore, a grouping of "concrete," "relatively concrete," and "abstract" shopping trips results in a more computationally efficient identification of shopping missions and enhances the interpretability of the results of the subsequent analysis. Providing larger cluster input for the subsequent phase sharpens the accuracy of deriving meaningful product category associations for the identification of shopping missions. Independent analyses for both the joint "relatively concrete"

Table 5
Trip Type Segmentation Results for all Stores.

Shopping trip type	Hypermarket						Supermarket						Convenience store					
	Pilot			Control			Pilot			Control			Pilot			Control		
	Variety range	Baskets	Value	Variety range	Baskets	Value	Variety range	Baskets	Value	Variety range	Baskets	Value	Variety range	Baskets	Value	Variety range	Baskets	Value
Very concrete	"1–3"	3%	1%	"1–3"	3%	1%	"1–2"	8%	4%	"1–2"	9%	4%	"1"	13%	7%	"1"	12%	6%
Relatively concrete	"4–11" "12–23"	49%	48%	"4–12" "13–24"	53%	46%	"3–9" "10–16"	75%	63%	"3–8" "9–18"	73%	61%	"2–5" "6–11"	80%	77%	"2–5" "6–13"	82%	79%
Abstract	"24–36" "37+"	48%	51%	"25–36" "37+"	44%	53%	"17–40" "41+"	17%	33%	"19–42" "43+"	18%	35%	"12–17" "18+"	7%	16%	"13–20" "21+"	6%	15%

group of baskets with 3 to 16 categories and the separated clusters of 3–9 and 10–16 categories did not produce different results in the identification of shopping missions for the pilot supermarket. Table 5 also provides the total revenue and total share of basket for the three levels of shopping goal abstractness, which is critical to assess the practical value of this study. The total share of baskets and total share of value for the relatively concrete shopping trips decrease with store size. For instance, the value of "relatively concrete" baskets decreases from 78% in the convenience stores to 62% in the supermarket stores under consideration. However, even for hypermarket stores, typically preferred for bulk and abstract shopping, 50% of shopping trips (in terms of both total share and value) are applicable to the identification of shopping missions.

5.5 . Shopping mission identification

For each store, a second clustering analysis pertains to each of the three shopping goal abstractness levels. The results for the baskets of "relatively concrete" abstractness reveal distinct and diverse shopping missions; the results for the baskets of "concrete" and "abstract" goals do not produce any interpretable outcomes. These findings lend support to the expectation that shoppers with relatively concrete (cf. concrete or abstract) shopping goals are more likely to operate under deliberative (implemental) mindsets. This conclusion ascribes to the balanced number of product categories that relatively concrete shopping goals involve, compared with the limited number of product categories that concrete or the large number of product categories that abstract shopping goals involve.

Therefore, the remaining analyses cluster, relatively concrete shopping trips, indicated by basket variety, on the basis of the product categories they contain. This process does not weigh the variables, because the product taxonomy preparation ensures that all derived product categories are at the same level of detail and thus of equal discriminative importance. The 75 derived category fields, representing the contents of each basket, provide the clustering input variables. The Jaccard index serves as a proximity measure for capturing (dis)similarity between baskets. The Jaccard index consists of the size of the intersection between two vectors, divided by the size of their union, and is well suited for data sets with binary values. The k-centroid version of k-means cluster analytic method, compliant with the Jaccard index distance measure, then applies. Specifically, the *kcca* function applies, as implemented in the *flexclust* package for the *R* software for statistical computing. Similar to the shopping goal segmentation phases, training and testing sets with 60% and 40% of observations can be selected, and the gap statistic procedure derives the optimal number of clusters. Table 6 presents the final cluster solution for the pilot supermarket, including basket descriptions and participation probabilities for the 25 most popular product categories in the data set. The final cluster solutions for the remaining stores in this study are available in the Web Appendix.

Table 6 suggests that of the total 412,786 baskets in the supermarket pilot store, 309,591 baskets with a basket variety between 3 and 16 unique categories can be segmented into seven clusters. Cluster 1,

Table 6
Final cluster solution for the identification of shopping missions for supermarket pilot (Fact Table 3).

		All	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5	Cluster 6	Cluster 7
Basket descriptives	Average basket variety (categories)	9.79	7.82	9.62	11.78	10.41	6.54	13.73	14.34
	Average basket size (items)	20.63	20.33	19.88	19.63	22.03	21.47	23.97	27.66
	Average basket value (€)	17.41	13.66	14.67	14.61	17.59	20.59	24.53	36.43
	Total share of baskets (%)	100%	27%	21%	15%	13%	11%	7%	6%
	Number of baskets	309,591	83,589	65,014	46,438	40,249	34,055	21,671	18,575
Product categories	Milk	150,204	100%	23%	26%	42%	38%	37%	27%
	Juice	119,471	75%	11%	43%	13%	13%	18%	47%
	Coffee and tea	103,805	68%	7%	13%	13%	12%	100%	21%
	Chocolate and confectionary	99,586	34%	7%	100%	13%	18%	19%	19%
	Eggs	95,654	39%	59%	6%	9%	34%	8%	18%
	Carbonated drinks	89,450	2%	43%	78%	9%	43%	3%	17%
	Biscuits and sweets	88,840	27%	11%	63%	16%	16%	73%	14%
	Packaged bakery	88,662	46%	13%	16%	16%	5%	21%	63%
	Shampoo and shower gel	80,085	6%	36%	45%	100%	14%	3%	16%
	Packaged cheese	73,616	6%	36%	0%	12%	24%	3%	84%
	Fresh vegetables	72,317	12%	72%	7%	8%	12%	15%	9%
	Cereals	68,964	63%	6%	8%	8%	7%	11%	6%
	Yoghurt	67,048	43%	10%	19%	12%	13%	22%	12%
	Fresh meat	65,494	5%	68%	7%	10%	21%	9%	8%
	Pasta	60,943	7%	13%	7%	16%	100%	17%	8%
	Condensed milk	55,903	27%	11%	9%	9%	2%	68%	12%
	Canned food	55,178	0%	29%	4%	6%	82%	10%	8%
	Freshly cut cheese	55,169	7%	56%	6%	6%	17%	6%	5%
	Freshly cut chartuterie	49,880	5%	3%	4%	8%	18%	13%	78%
	Chips and snacks	46,900	6%	8%	52%	13%	8%	18%	9%
	Household cleaning	44,163	9%	12%	8%	100%	14%	7%	5%
	Rice	44,541	4%	6%	11%	14%	72%	8%	9%
	Dry fruits and nuts	38,537	5%	3%	63%	10%	1%	3%	0%
	Detergents and washing	38,141	9%	6%	8%	88%	9%	10%	4%
	Fresh fruit	33,039	9%	9%	17%	23%	4%	7%	11%
	Oral care	21,767	5%	4%	4%	38%	4%	7%	8%
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involving 27% of the total share of baskets, is characterized by high participation probabilities for milk, juice, coffee and tea, and cereals, referring to a "breakfast" shopping mission. A "meal" shopping mission is apparent in the 21% of baskets included in Cluster 2, dominated by categories such as fresh meat, eggs, and freshly cut cheese. Chocolate and confectionary, carbonated drinks, and biscuits and sweets recur in the "snack" shopping mission baskets of Cluster 3, whereas categories such as shampoo and soaps, household cleaning, and detergents and washing products are in baskets bundled around the "cleaning" shopping mission of Cluster 4. Pasta, canned food, and rice product categories characterize baskets grouped in Cluster 5, revealing a second "meal" shopping mission; coffee and tea, biscuits and sweets, and condensed milk characterize baskets grouped in Cluster 6. Cluster 7 features a large proportion of cheese and charcuteries that refer to "hors d'oeuvres," pre-meal (or post) snacks. The category distributions across shopping missions reveal that generally popular categories (e.g., milk and biscuits, sweets; Table 6) can serve as cross-selling "bridges" across shopping missions. Furthermore, categories that score high in some missions (e.g., eggs, which appear 59% in Cluster 2, canned food 82% in Cluster 5) but have growth potential in others (eggs 39% in Cluster 1, canned food 29% in Cluster 2).

This evidence supports both the replicability and the validity of the results (Pastor, 2010). Replicability exists according to the proportions of shopping missions identified when independently analyzing the data from the pilot and control stores with the same size (format) and clientele geodemographic properties. In support of validity, the variables exogenous to the cluster model relate to the results in ways anticipated by the theoretical development, previous research, and the exploratory focus group discussions (Breckenridge, 1989). Shopping missions identified in smaller stores are of significantly smaller size and value compared with those of larger stores. Smaller stores preferred by consumers for smaller shopping trips predominately serve shopping missions for daily essentials, such as "snacks" and "breakfast." Larger stores with extensive product assortments, better prices, and parking facilities instead attract larger shares of the "meal" and "cleaning" shopping missions, associated with expensive, voluminous products. Fig. 4 summarizes the composition of the shopping missions identified for the six stores, along with the top three product categories for each shopping mission.

6 . Discussion

This research show that novel technologies, in addition to introducing new shopper touch points, offer advanced analytical capabilities that can indicate how to best use the sheer amount of data already available and tap their shopper marketing potential. Building on shopping goal abstractness and mindset theory literature (Bell et al., 2011; Gollwitzer, 1990; Lee & Ariely, 2006), this article conceptualizes shopping missions as different instantiations of, or values assigned to, shopping goals under a deliberative mindset. Furthermore, this research employs the design science paradigm (Hevner et al., 2004) to present an end-to-end analytical method that informs the composition of the shopping missions for different stores and formats. This method shows that the identification of shopping missions is contingent upon and an antecedent of, shopping trip abstractness and shopping mindset. Furthermore, the process of designing measures for abstract purchasing behavioral constructs, such as trip types and shopping missions, is readily available through the exploitation of sales data.

These results reveal that trips of relatively concrete abstractness constitute a large portion of store visits. However, even if the proportion of relatively concrete shopping trips decreases for larger store formats, their importance in terms of total share of baskets and revenue generated remains high (50% for hypermarkets). Six predominant shopping missions emerge for the stores under consideration: "breakfast," "meal," "snack," "cleaning," "coffee," and "light meal & hors d'oeuvres." To discriminate and interpret the cluster analysis results and identify shopping missions, the authors suggest using category participation probabilities for each cluster and reporting differences among the mean size, variety, and value descriptive attributes of shopping missions. Finally, in line with past literature, the tacit channel-category associations, focus group discussions, and statistical learning analyses suggest patterns of format–mission relationships, such that larger formats accommodate "meal" and "cleaning," whereas convenience stores mostly support "snack" and "breakfast" shopping missions. Knowing the total share of baskets for each mission, practitioners can develop format- or even store-specific, mission-informed shopper marketing actions, uncover insights to strategically position brands and categories within specific shopping missions, and better understand the conditions in which

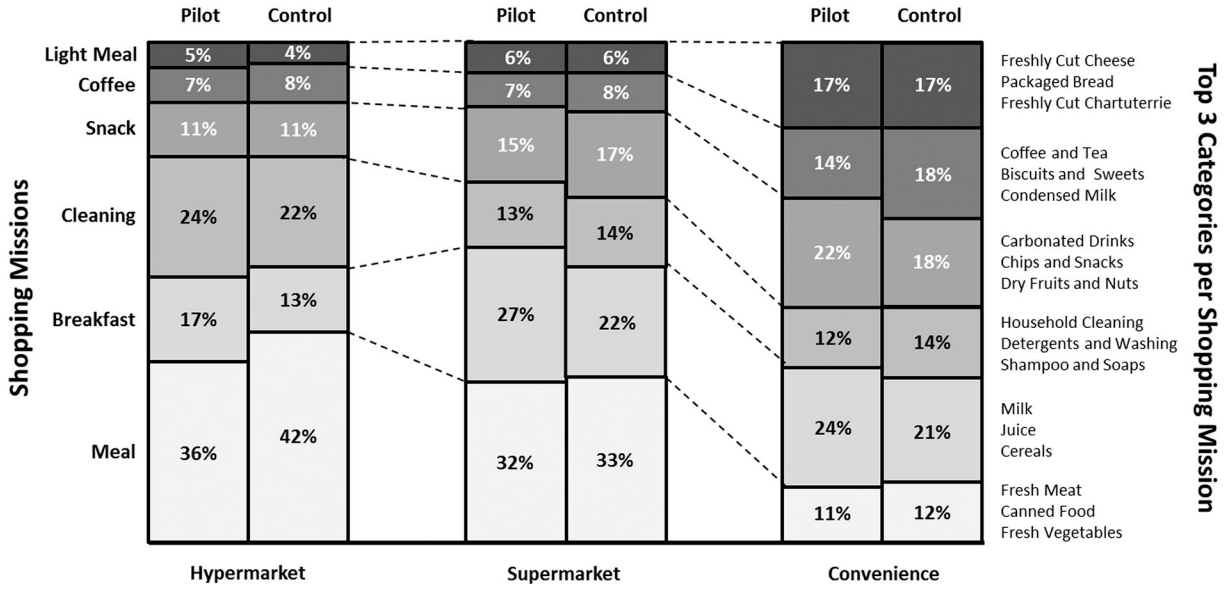


Fig. 4. Composition of shopping missions across stores.

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certain trips occur, relate to money spent, and influence coupon redemption and pricing decisions.

The identification of shopping missions also addresses recent calls from both academia and practice to move the focal points beyond standalone brands and isolated categories to shopper marketing activities across categories (ECR Europe, 2011; Russell & Petersen, 2000). Furthermore, data-driven approaches in shopper behavioral analysis provide room for innovation in retail business models (Chesbrough, 2010). Even with limited data sources and minimum investments in infrastructure and business processes, retailers can leverage their distinct advantage of being the customer's closest link to the marketplace (Lusch, Vargo, & O'Brien, 2007), establishing themselves as infomediaries between shoppers and manufacturers of consumer packaged goods (Grover & Teng, 2001).

6.1 . Limitations and further research

Additional research should address some limitations of this study and build on its findings. Shopper marketing research for theory development can test whether the identified instantiations of the shopping mission construct, such as "meal," "breakfast," or "snack," interact with deliberating processes involving in-store decisions among numerous substitutive or supplemental grocery categories. In addition, it would be interesting to investigate whether both shopping trip abstractness and mindset are assumed prior to entering the store and whether the shopping mission antecedent variable also serves as a preshopping factor that affects in-store behavior and decision making. Experiments with how shoppers respond to the presence or absence of mission-framed shelf assortments, floor space organizations, and promotions may be particularly interesting. Further research investigating situations of cognitive matching or conflicting shopping missions and their impact on the processing fluency of in-store information and experience also would be insightful. What if a shopper enters with a "breakfast" shopping mission and comes across a "breakfast-relevant" promotion involving a free gift versus a "cleaning" special offer for 20% off? Finally, research should test whether grocery shopping trips performed through computer-mediated environments, such as retail websites, are more targeted and concrete and consciously refer to predetermined shopping goals.

Research adhering to the design science paradigm also could improve the contributions of this study by striving to design and test a fully implementable, scalable, system instantiation of the proposed method that allows for ongoing analyses and reports of results required in real-life settings. Furthermore, research on how to complement this approach with customer-unique identifiers and demographic data, as found in loyalty schemas, would improve the insightfulness of the method. An interesting approach would be to know how shopping missions vary and cross-tabulate them across gender, age groups, and RFM (recency, frequency, and monetary value) database groups. Further research could also shed light on the whether shopping missions vary across regional geodemographic properties; do urban stores accommodate different shopping missions than rural stores, for example? Customer-unique identifiers could reveal changes in the composition of shopping goals and shopping missions across time. Customer-specific timestamps may reveal the frequency of each shopping type and mission or the time of day or week when each is most likely to occur. Practitioners could use this information to add samples and displays during these time periods to highlight complementary products or recipes and solutions. Research also could examine whether, during the product taxonomy preparatory step, retailers can derive both meaningful category descriptions and estimates of profit margins per category. For example, by computing the weighted average of the margins of all the SKUs of a category, retailers could estimate the relative importance of each shopping mission in terms of total share of baskets, revenue, and profit.

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